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Detecting stock market regimes from option prices

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Abstract

Equity market returns alternate between periods of calm and crises. Researchers commonly employ regime switching models to capture this behaviour. We show that forward-looking information extracted from option prices improves regime detection. In particular, horizon spreads in option-implied equity risk premia allow earlier detection of regime switches and improved prediction of the equity premium. This findings holds across recent disaster periods like the 2008/2009 financial crisis and the 2020 Covid pandemic outbreak, in both US and Emerging equity markets.

JEL Codes: G01, G12 Options, Regime switching, Equity risk premium

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1 Introduction

Asset prices observed in the stock market often exhibit abrupt changes. While some changes can be momentary (such as jumps), other changes can persist for a longer period. For example, the means and volatility of stock market returns during the global financial crisis in 2007-2009 were distinctly different in the period before and after the crisis. Regime switching models are well suited to capture these changes that are more persistent over time.

However, since most applications of regime switching models focus on observed returns, researchers ignore potentially useful information such as investors' risk preferences and beliefs that can be extracted from option prices. Equity index options for example, are among the most widely used instruments for risk management. These options integrate forward-looking information in a timely manner. This paper draws on information extracted from equity index option prices to sharpen regime detection in equity markets.

To be more specific, we extract the long term and short term expected equity risk premium from equity index options. We use the difference between the long and short term premium, the *horizon spread*, as a proxy for the market state. We then apply a hidden Markov model to detect shifts in stock market regimes during a sample that includes both the 2008-2009 financial crisis and the 2020 Covid pandemic. Applying a regime switching model to the horizon spread has distinct advantages over more standard approaches, such as estimating a regime switching model using observed returns or conditional volatility (GARCH).

Regime switching behaviour of stock returns is intuitive and the concept has been widely used to model financial time series (see Dias et al. [10] and Zheng et al. [17]). Regime switching models are well suited to model several conditional distributions nested in a single financial time series. The mixing of multiple conditional distributions produces important non-linear effects, such as excess kurtosis, volatility clustering, and time-varying correlations that are typical for financial time series. Regime switching models distinguish the different

distributions and attribute each of them to a relevant regime. In doing so, such models capture the non-linearities of financial time series in a parsimonious manner.

Moreover, regime switching models for financial time series have important implications in both asset pricing and asset allocation decisions. For example, Capponi and Figueroa-Lopez [7] propose an optimal portfolio allocation model with regime switching considerations, while Elliot et al. [11] and Capponi et al. [8] built on regime-switching models to price options. More precisely, [11] uses a regime switching model with feedback effect to price options, while [8] uses the regime switching model to price path dependant claims, such as bonds or barrier options.

The common regime switching model used in fitting financial time series is built on the structure:

$$y_t = \mu_{s_t} + \phi_{s_t} y_{t-1} + \sigma_{s_t} \varepsilon_t \qquad \varepsilon \sim iid(0,1), \quad s_t \in \{0, 1, \ldots, k\}$$

where s_t is the regime at time t, y_t is the parameter in question, such as the stock market returns, and σ_{s_t} is the volatility of y. This structure captures the influence of regimes on the unconditional mean μ_{s_t}, the auto-correlation ϕ_{s_t} and the volatility σ_{s_t}. Such models are typically calibrated from historical returns observations [2].

We take a different approach and try to detect regimes from forward-looking information. More specifically, instead of looking at the observed stock market returns, we use the option implied expected equity premium to estimate the underlying regimes of the stock market. Our study is complementary to the work of [11] and [8]: while they use regime-switching models to price options, we attempt to detect switching regimes from option prices.

Options written on equity and equity indexes are among the most commonly used derivatives. Such options are widely traded and their prices reflect the expectations of investors on the underlying asset. In particular, variations in volatility play a key role in options pricing (see Da Fonseca et al. [9]). The flip-side is that option prices provide useful information on investor expectations regarding volatility and - more generally - the distribution of asset returns. We use information from option markets to extract investor expectations about

the underlying asset. A different use of option markets would be to consider augmenting the asset space by adding options to the investor's asset menu (see Jewell et al. [13]). The information we focus on is information concerning expected returns on the aggregate equity market index. In a recent study, Martin [14] shows that the expected equity risk premium can be estimated from equity index options. More precisely, one can estimate investors' required rate of return for the underlying index over the time period until the option expiry date. Since equity options are created with a range of expiry dates ranging from a week to one year, the required return can be extracted for different time horizons. This feature provides an additional dimension of information about investor's expectations, as we can compare the required return for different time horizons.

In the classic asset pricing models, such as the habit formation model of Campbell and Cochrane [5] and the long-run risk model of Bansal and Yaron [4], the long term equity risk premium should be higher than the short term equity risk premium. Following the reasoning of these models, investors should require a higher long term return to compensate for the long-run risks, as compared to holding the asset in the short term.

However, during a crisis, the imminent risk can be extremely high, and exceed the risk in the long-run, i.e. after the crisis is over. Under this condition, the investor's short term required return (to compensate the short term risks) may well exceed the required return in the long term. In fact, Aït-Sahalia et al. [1] show empirically that, although the *unconditional* long term equity risk premium is higher than the short term equity risk premium, the *conditional* short term premium becomes higher than the *conditional* long term premium over crisis periods.

Aït-Sahalia et al. [1] focus on observed returns, where the observed equity risk premium are constructed by holding the equity index and the index dividends. We constructed a different measure of horizon effects using the expected equity risk premium estimated from equity index options. Using equity index options, we extract the short term expected risk premium from options with short time-to-expiry, and the long term expected risk premium

from options with longer time-to-expiry. We define the horizon spread Δ_{HS} as the difference between the long term and the short term expected returns, and use this horizon spread as an indicator to detect regime shifts in the stock market. Similar to the findings from historical returns, we expect the horizon spread to be positive in normal times, but to turn negative during a crisis.

Our study shows that the option implied horizon spread performs better than observed returns in regime detection. In particular, using horizon differences in the option-implied equity risk premium allows earlier detection of regime switches and produces clear regime switching probabilities. This findings holds across recent disaster periods like the 2008-2009 global financial crisis (GFC) and the 2020 Covid pandemic outbreak, in both US and emerging equity markets.

To illustrate the practical implications of the improvements in regime detection, we use the alternative regime switching indicators in a forecasting exercise. More precisely, we attempt to forecast the future equity risk premium using a step-wise model selection approach under different equity market regimes (Guidolin and Pedio [12]). In this application, we find that the model based on option implied horizon spread produces better forecasting performance than a set of competing models (based on observed returns) that are less efficient in regime detection.

The following section describes the methodology to extract the expected equity premium from options and reviews the model used to estimate the switching regimes. Sections 3 and 4 present the data and the empirical results. Section 5 reports implications of our findings in a forecasting application.

2 Methodology

2.1 Extracting the expected equity risk premium from options

Martin [14] shows that the expected equity risk premium can be expressed as the equation:

$$\mathbb{E}_t R_T - R_{f,t} = \frac{1}{R_{f,t}} var_t^* R_T - cov_t(M_T R_T, R_T) \tag{1}$$

where $\mathbb{E}_t R_T - R_{f,t}$ is the expected equity risk premium at time t for the time horizon T, R_f is the risk free rate, $var_t^* R_T$ is the conditional risk neutral variance of R_T, and M_T is the stochastic discount factor for time T.

To proxy for the equity risk premium, we take R_T as the gross return of a broad equity market index, such as the S&P 500. Since the aggregate stock market, as proxied by the equity index, is a non-hedging asset, the second term in equation 1 should be negative, i.e., $cov_t(M_T R_T, R_T) \leq 0$. We can then rewrite Equation 1 as :

$$\mathbb{E}_t R_T - R_{f,t} \geq \frac{1}{R_{f,t}} var_t^* R_T$$

implying that $\frac{1}{R_{f,t}} var_t^* R_T$ is the lower bound of the expected equity risk premium. Using US options data, Martin [14] and Martin and Wagner [15] show that the lower bound $\frac{1}{R_{f,t}} var_t^* R_T$ is sufficiently tight, making it a good proxy for the expected equity risk premium.

As a risk-neutral measure, the conditional risk neutral variance of R_T can be readily estimated from options data. Following Martin [14], we estimate the conditional risk neutral variance non-parametrically from both put and call prices :

$$\frac{1}{R_{f,t}} var_t^* R_T = \frac{2}{S_t^2} \left[\int_0^{F_{t,T}} put_{t,T}(K) dK + \int_{F_{t,T}}^{\infty} call_{t,T}(K) dK \right]$$

As discussed in section 2.1, we are interested in the difference between the long term and

short term expected equity risk premium. We compute the horizon spread $\Delta_{HS_{i,t}}$ as below:

$$\Delta_{HS_{i,t}} = E\left[R_{i,t}\right]_L - E\left[R_{i,t}\right]_S$$

Where i indicates the relevant market, $E\left[R_{i,t}\right]_L$ is the long term expected risk premium extracted from options with long time-to-maturity $T - t = L$ and $E\left[R_{i,t}\right]_S$ is the short term expected risk premium extracted from options with short time-to-maturity $T - t = S$, and $L > S$.

2.2 Hidden Markov regimes in the stock market

To estimate the hidden Markov regimes present in the stock market, we consider time series data denoted by $Y_{k:T} := (Y_1, Y_2, \ldots, Y_T)$ for the length of time T and state k. We further denote the state variables as $S_{k:T} := (S_1, \ldots, S_T)$, where the state variables S_t are random with discrete values that come from a finite set $\mathcal{S} = 1...n$, as $S_t = k, k \in \mathcal{S}$. The set $\mathcal{S}$ is the state-space of the hidden Markov model, and n is the number of states of the model. The observations Y_t are dependent on the state variables S_t such that the distribution of Y_t can be written as: $f_k(Y_t) := f(Y_t|S_t = k)$. As $\mathcal{S}$ is a finite set, this implies that the probability distribution of the data Y_t is a mixture distribution with n components, and each component corresponds to the states $\mathcal{S}$ of the model:

$$f(Y_t) = \sum_{k=1}^{n} p_k f_k(Y_t)$$

where p_k is the proportion of each component (state) k, subject to $\sum_{k=1}^{n} p_k = 1, p_k \geq 0$, $f_k(\cdot)$ is the conditional probability distribution of Y_t under state k.

We assume the current state follows a Markov process and depends only the previous state:

$$P(S_t|S_1, \ldots, S_{t-1}) = P(S_t|S_{t-1})$$

Under this assumption, the transition probabilities of each regime can be written as:

$$a_{jk}(t) := P(S_{t+1} = j | S_t = k), \qquad j, k = 1, \ldots, n.$$

The joint likelihood of the time series data and the hidden state sequence of the model can then be estimated as :

$$L(Y_{1:T}, S_{1:T} | \lambda) = \pi_{S_1} f_{S_1}(Y_1) \Pi_{t=1}^{T-1} a_{S_t} S_{t+1} f_{S_{t+1}}(Y_{t+1})$$

where λ is the parameter vector of the hidden Markov model. λ consists of three different components: the initial state probabilities π, the transition model which determines the transition probabilities A, and finally the the state dependent probability distributions $f_k(Y_t)$.

Model Specification

We follow Dias et al. [10] and assume that there are two hidden states ($n = 2$), an expansion state and a contraction state. We estimate the hidden Markov model while setting $Y_t = \Delta_{HS_{i,t}}$ the horizon spread. We also assume that the horizon spread $\Delta_{HS_{i,t}}$ follows a Gaussian probability distribution for all i.

For comparison, we also estimate regime shifts using the observed returns and the conditional volatility of the equity market index. In particular, we use the Markov switching GARCH model of Ardia et al. [3] to detect regimes using the conditional volatility extracted from observed stock market returns.

We now turn to section 3 to present the data, before discussing results in section 4.

3 Data

Our objective is to detect the regime shifts in the US and emerging stock markets using equity index options data. We used daily options data provided by CBOE Global Markets

and OptionMetrics. The one-month US TBill rate (proxy for the risk free rate) and index dividend yield are collected from Refinitiv Datastream. More specifically, we study the expected equity risk premium of the US market using options written on the S&P 500 index, and of emerging markets using options written on the MSCI Emerging Market index (EEM). The MSCI emerging market index includes 26 countries, with three East Asian countries (China, Taiwan and South Korea) dominating the index (up to 70% weight in 2020). We chose to study the options traded on the EEM index, as they are the most widely traded options for emerging markets. In addition, the three dominant East Asian markets in the EEM index are also the first countries to experience the Covid pandemic, months before North America and Western Europe were affected [16]. Thus, the emerging markets might reveal an earlier shift in regime compared to the US markets during the Covid pandemic.

Our sample period starts in January 2006 and ends in August 2020, covering both the global financial crisis (GFC) in 2007-2009, as well as the beginning of the Covid pandemic from December 2019 to the summer of 2020.

To avoid the undue influence of illiquid options, we omit replicated entries and observations where the closing bid price is zero or absent. If the option is traded on more than one exchange, we use data from the primary exchange. In addition, options with less than 7 calendar days to expiry are excluded. Table 1 shows the remaining number of options after filtering. For example, there are 20 million options traded on the S&P 500 index in our option universe, where 15.8 million of these options expire within 6 months (between 8 to 180 days), another 2.2 million options expire within 6 to 12 months, and the remaining 2 million options have more than one year to expiry.

Table 1: Options traded on US and emerging stock market indexes

The table below presents the total number of S&P 500 and MSCI Emerging Market (EEM) index options available during the sample period of this study. The information is grouped by time-to-expiry measured on the trading day, based on daily data extracted between the time period from January 2006 to August 2020.

Time to expiry in days	$8 \leq \tau < 180$	$180 \leq \tau < 360$	$360 \leq \tau$	Total
United States	15,769,528	2,290,156	2,019,155	20,078,839
Emerging Markets	1,651,091	726,969	322,146	2,700,206

On each trading day, the expected equity risk premium is estimated for different horizons, using the procedure from section 2.1. For the discrete integration to work, the expected risk premium will only be computed if there are more than 3 put and call options for a particular time horizon on the trading day. The range of time horizons extracted on each trading day varies, as the time-to-maturity shortens closer to the expiration dates of the options. In order to extract the expected risk premium for the fixed set of time horizons at T=30, 90, 180 and 360 days on a daily basis, these risk premia are extracted using linear interpolation of the bounds computed for the available set of time horizons of each trading day.

Table 1 indicates that the number of options traded for more than 300 days of maturity is relatively low. Consequently, we compute the expected equity risk premium for up to 180 days. We use the 30 days expected equity risk premium as the short term premium $E[R]_S$, and the 180 days expected risk premium as the long term premium $E[R]_L$.

Descriptive Statistics

Leading asset pricing models featuring habit formation [5] or long run risks [4] stipulate that the expected returns of investors increase with a longer time horizon. Table 2 shows descriptive statistics for the expected risk premium for the US and emerging stock markets. Panel A of this table shows that our results are broadly in line with this prediction. The median value of the expected risk premium is higher for the long term than for the short term, both in the US and in Emerging Markets. We also observe that the expected risk premium is almost twice as high for emerging markets (6.3%) as it is for the US market (3.3%). This is not surprising, as Emerging Markets exhibit higher political risk and higher volatility (3.5% in Emerging Markets versus 2.8% in US) and thus warrant higher required returns from investors.

Panel B and C of table 2 report descriptive statistics of the equity risk premium during the financial crisis and the Covid 19 pandemic; during such disaster periods, we observe an increase in both the volatility and the expected risk premium, as expected. We also see from

Table 2: Descriptive Statistics of the Expected Equity Risk Premium

The table below presents the descriptive statistics of the expected risk premium and regime switching indicators for the US and emerging stock markets. The equity risk premium and regime switching indicators are reported based on the data extracted between the time period from January 2006 to August 2020. Returns reported are annualised, the conditional volatility is estimated with a EGARCH(1,1) model.

	Mean	**SD**	**Median**	**Min**	**Max**	**Skewness**	**Kurtosis**
	Panel A : Full Sample (2006 - 2020)						
US							
Short term	3.28%	2.83%	2.21%	0.90%	44.54%	2.312	5.789
Long term	3.59%	2.08%	2.76%	1.23%	20.48%	1.653	2.499
Returns	8.18%	16.59%	0.06%	-5.02%	4.38%	-0.362	3.528
Cond. Vol.	14.98%	7.72%	0.79%	0.38%	2.91%	1.377	1.665
Horizon Spread	0.36%	1.08%	0.61%	-5.85%	1.81%	-2.743	9.349
Emerging Markets							
Short term	6.29%	3.45%	5.38%	1.99%	89.55%	1.120	0.535
Long term	6.28%	2.61%	5.51%	2.95%	49.41%	0.910	-0.108
Returns	7.93%	25.52%	0.10%	-7.42%	7.01%	-0.180	2.530
Cond. Vol.	21.44%	6.52%	1.27%	0.69%	2.56%	0.880	0.295
Horizon Spread	0.00%	2.23%	0.26%	-12.98%	1.98%	-2.478	7.377
	Panel B: Financial Crisis (2007-2009)						
US							
Short term	8.79%	8.12%	5.45%	0.94%	44.54%	1.919	3.837
Long term	6.24%	4.26%	4.91%	1.23%	20.48%	1.175	0.488
Returns	-25.01%	26.30%	0.02%	-5.02%	4.34%	-0.076	0.589
Cond. Vol.	27.40%	14.95%	1.50%	0.49%	4.78%	1.268	1.089
Horizon Spread	-2.05%	4.63%	-0.71%	-31.72%	2.73%	-3.069	12.856
Emerging Markets							
Short term	22.36%	18.14%	15.33%	5.42%	89.55%	1.940	3.377
Long term	16.65%	10.22%	12.57%	6.39%	49.41%	1.396	0.960
Returns	-30.35%	40.72%	0.08%	-7.42%	7.01%	-0.180	0.388
Cond. Vol.	42.31%	22.87%	2.25%	0.94%	7.60%	1.530	2.147
Horizon Spread	-6.05%	9.33%	-2.47%	-44.14%	1.61%	-2.434	5.669
	Panel C: Covid 19 (Dec 2019-2020)						
US							
Short term	8.79%	8.97%	6.55%	1.30%	45.10%	2.098	4.656
Long term	6.42%	3.64%	6.76%	2.02%	18.84%	0.683	0.489
Returns	20.73%	29.01%	0.17%	-7.85%	9.96%	0.402	2.864
Cond. Vol.	27.00%	17.69%	1.63%	0.35%	4.82%	0.826	0.103
Horizon Spread	-2.28%	5.76%	0.16%	-28.57%	1.64%	-2.496	5.992
Emerging Markets							
Short term	11.34%	13.55%	7.09%	2.11%	85.96%	3.368	13.189
Long term	7.41%	4.06%	7.26%	2.97%	25.59%	1.778	5.070
Returns	11.63%	29.15%	0.28%	-11.42%	7.41%	-1.105	5.489
Cond. Vol.	29.61%	19.42%	1.48%	0.72%	6.61%	2.082	3.844
Horizon Spread	-3.18%	7.61%	-0.25%	-44.63%	2.07%	-3.039	10.472

these results that there is a reversal in the horizon effect of the equity risk premium during disasters. In both the US and emerging markets, investors' expected risk premium over the short term is higher than that over the long term during these crisis periods. For example, during the Covid pandemic, the long term expected risk premium in Emerging Markets is 7.4%, which is lower than the short term expected risk premium at 11.3%. Likewise, in the US data, the long term premium during the Covid pandemic is only 6.4%, while the short term premium is 8.8%.

These results suggest that investors expectation changes over crisis periods. The imminent risk during a crisis is exceptionally high, prompting investors to require higher returns in the short term. On the contrary, the required return decreases in the longer horizon, as the investors expect the uncertainty to abate over time. This observation contradicts the prediction of the major asset pricing models that the risk premium will increase with longer time horizon.

We compute the difference between the long term and short term expected risk premium as the horizon spread Δ_{HS} and use it as an indicator to detect regime shifts in the stock market. We expect the horizon spread Δ_{HS} to be positive in normal times (as stipulated in the major asset pricing models), turning negative during times of crises.

Table 2 reports the descriptive statistics of the horizon spread. The horizon spread Δ_{HS} is non-negative over the full period for both regions, but turns negative (-2.1% US, -6.1% emerging markets) on average during the Financial crisis, and over the Covid 19 pandemic (-2.3% US, -3.2% emerging markets).

4 Results

This paper proposes to estimate regime switching models for the option implied risk premium. For comparison with our approach, we also estimate regime switching models using historical returns and conditional volatility of the equity index.

Figure 1: Boxplots of Regime Switching Indicators for United States

Figure below presents box plots for the three regime switching indicators in the US market. The range of the whiskers are winsorized at 95%. The information is presented for the full period based on the data extracted from January 2006 to August 2020, as well as for the two sub-periods: Global Financial Crisis (GFC) from 2007-2009 and the first wave of the Covid pandemic from December 2019 to the summer of 2020.

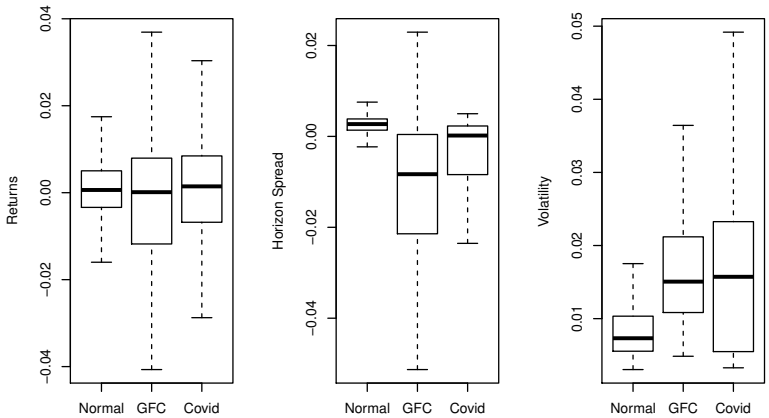
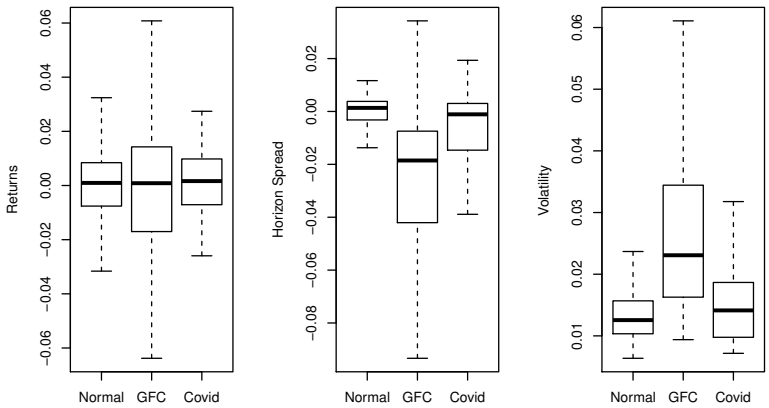


Figure 1 shows box plots for the three regime switching indicators in the US market. For each indicator (historical returns, the horizon spread, and conditional volatility), figure 1 shows a box plot to summarise the distribution during normal times, and box plots summarising the distribution during the Global Financial Crisis and the Covid-19 outbreak. Figure 2 shows the same information for Emerging Markets. The leftmost plot in both figures shows that the median historical returns are very similar during the different periods. While, the range of observed returns over the global financial crisis (GFC) and Covid periods are larger than during normal times, the median is almost indistinguishable across the three sub-samples. The right most plot for conditional volatility shows a very different picture. Over the crisis periods, the median and range of the conditional volatility estimates are clearly higher than during normal times. This suggests that conditional volatility may be a more suitable indicator for detecting regime switches than observed returns.

The plot in the middle of Figures 1 and 2 shows the distribution of the horizon spread Δ_{HS} across the sub-samples. The plots show that the median of the horizon spread in normal times is markedly different from its median during crisis periods. In particular, the

Figure 2: Boxplots of Regime Switching Indicators for Emerging Markets

Figure below presents box plots for the three regime switching indicators in the emerging markets. The range of the whiskers are winsorized at 95%. The information is presented for the full period based on the data extracted from January 2006 to August 2020, as well as for the two sub-periods: Global Financial Crisis (GFC) from 2007-2009 and the first wave of the Covid pandemic from December 2019 to the summer of 2020.



median horizon spread is positive during the normal period, but turns negative over the crisis periods.

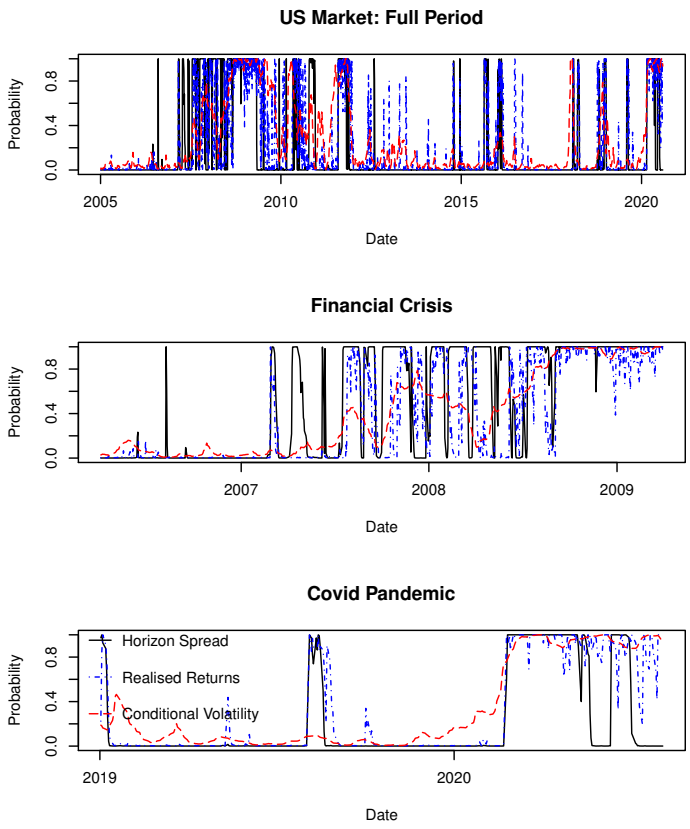
We proceed to an analysis of Markov regime models to detect regime switches in the stock market, as specified in section 2.2. We compare the results when using each of the three different indicators, namely historical returns, conditional volatility and the horizon spread. The results and estimated parameters for the US and emerging stock markets are presented in Figures 3, 4 and Table 3.

Figures 3 and 4 show the probability over time that each stock market enters the contraction phase. The figures show that in general, all three indicators are able to detect the contraction phase during the 2007-2009 financial crisis, as well the Covid pandemic in 2020. Of the three indicators, the conditional volatility tends to suggest a gradual switch from one regime to another, while the horizon spread provides the sharpest changeover, with probabilities that are either close to zero or one.

Inspection of the time series of contraction probabilities during the Covid pandemic

Figure 3: Hidden Markov States in the United States Stock Market

Figure below presents the probability over time that US stock market enters the contraction phase. The information is presented for the full period based on the data extracted from January 2006 to August 2020, as well as for the two sub-periods: Global Financial Crisis (GFC) from 2007-2009 and the first wave of the Covid pandemic from December 2019 to the summer of 2020.

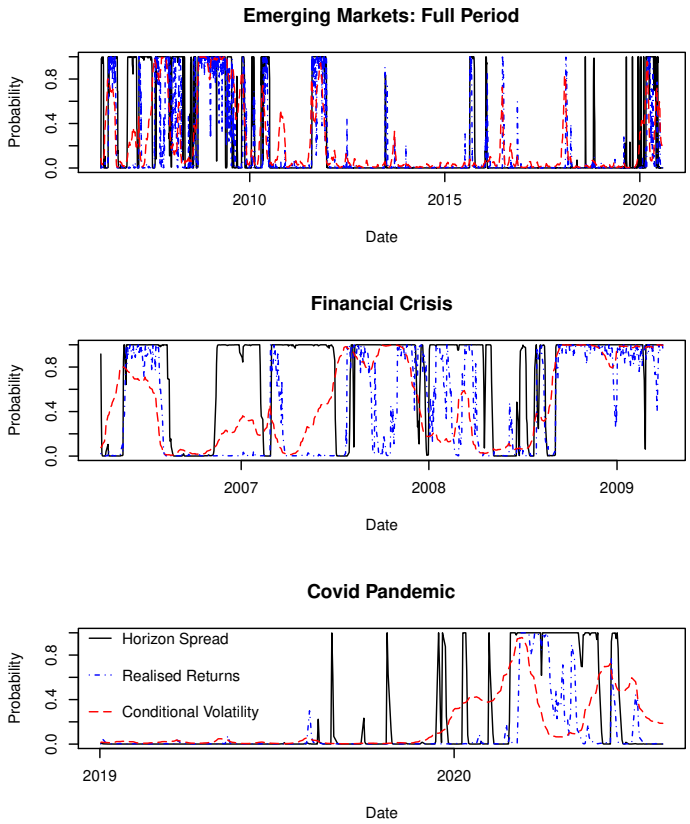


suggests that the horizon spread (in solid black) detects regime shifts earlier than observed stock market returns (in dotted blue), or conditional volatility (in red dashes). This is particularly clear in the case of the emerging markets. Figure 4 suggests that the horizon spread allowed to detect a contraction regime as early as December 2019, while historical returns did not detect a regime shift until March 2020. Conditional volatility also indicated an increasing probability that emerging markets were entering a contraction regime starting from December 2019, but the probability only attained a decisive level around March 2020.

The time series plots of contraction probabilities resulting from the three different indicators suggest that the horizon spread provides a sharper distinction between expansion

Figure 4: Hidden Markov States in the emerging Stock Market

Figure below presents the probability over time that emerging country stock markets enters the contraction phase. The information is presented for the full period based on the data extracted from January 2006 to August 2020, as well as for the two sub-periods: Global Financial Crisis (GFC) from 2007-2009 and the first wave of the Covid pandemic from December 2019 to the summer of 2020.



and contraction phases that historical returns or conditional volatility. To further explore this point, we present additional analysis of Regime Switching models using the different indicators in Table 3.

When the Markov regimes are extracted using the observed stock market returns, up to 16% (11.6% for emerging markets) of the state probability lies in the *indecisive gray area* (probability value between 0.1 to 0.9). Similarly, when the regimes are extracted using conditional volatility, about 34.1% (26% for emerging markets) of the state probability lies in the indecisive gray area. However, when the horizon spread is used as an indicator, only 4.6% (3.8% for emerging markets) of the state probability falls in the range between 0.1

Table 3: Parameters Extracted from the Markov Regime Models

The table below presents parameters extracted from the Markov Regime Switching models using the different indicators. The Markov models are estimated based on the indicators extracted from January 2006 to August 2020. The transition matrix describes the probability that one regime changes to another, where S1 is the expansion regime and S2 is the contraction regime. The probability distribution depicts the periods of *certitude* of the Markov models for predicting a contraction regime. Periods (measured by percentage of occurrence) with a probability of $p < 0.1$ and $p > 0.9$ are clear indications that the stock market is entering a contraction phase, whereas periods with the probability range between 0.1 and 0.9 are less clear if the stock market will enter a contraction phase.

Panel A: United States									
	Horizon Spread			Stock Index Returns			Conditional Volatility		
Transition matrix		to S1	to S2		to S1	to S2		to S1	to S2
	from S1	0.989	0.011	from S1	0.988	0.012	from S1	0.990	0.010
	from S2	0.052	0.948	from S2	0.043	0.957	from S2	0.041	0.959
Probability distribution	$p < 0.1$	$0.1 \leq p \leq 0.9$	$p > 0.9$	$p < 0.1$	$0.1 \leq p \leq 0.9$	$p > 0.9$	$p < 0.1$	$0.1 \leq p \leq 0.9$	$p > 0.9$
	79.8%	4.6%	15.6%	72.8%	16.0%	11.2%	56.1%	34.1%	9.9%
AIC		-32752			-25375			-26276	
BIC		-32708			-25331			-26207	
Panel B: Emerging Markets									
	Horizon Spread			Stock Index Returns			Conditional Volatility		
Transition matrix		to S1	to S2		to S1	to S2		to S1	to S2
	from S1	0.99	0.01	from S1	0.993	0.007	from S1	0.993	0.007
	from S2	0.029	0.971	from S2	0.034	0.966	from S2	0.034	0.966
Probability distribution	$p < 0.1$	$0.1 \leq p \leq 0.9$	$p > 0.9$	$p < 0.1$	$0.1 \leq p \leq 0.9$	$p > 0.9$	$p < 0.1$	$0.1 \leq p \leq 0.9$	$p > 0.9$
	72.1%	3.8%	24.2%	77.8%	11.6%	10.6%	60.1%	26.0%	6.3%
AIC		-24429			-20039			-20475	
BIC		-24386			-19995			-20407	

to 0.9. Using the horizon spread thus provides for markedly sharper identification of stock market regimes than using historical returns or conditional volatility. This result suggests that using forward-looking information in equity index option prices adds value to using the prices of the underlying index alone.

We also report the AIC and BIC information criterion for the Markov regime models using horizon spreads, observed stock market returns and conditional volatility respectively in Table 3. Both the AIC and BIC criterion suggest that the Markov regime model using the horizon spread is the preferred model for both the US and emerging markets.

The good performance of the horizon spread model is likely due to the more distinct bi-modal characteristic of the horizon spread distribution, as we have shown in Figure 1 and 2, as the underlying means detected for the expansion state and contraction state are clearly different (0.003 and -0.016) for the horizon spread estimates. While the conditional volatility also shows distinct means for the expansion and contraction states, due to the wide inter-quartile range, the overlapping in the range of conditional volatility values across various states are more pronounced than in the case of the horizon spread. The overlapping range of values render it less efficient to use conditional volatility as an indicator to detect shifts in regimes. Likewise, the regime models using observed returns performs the least well, as the underlying means are almost adjacent (0.001 and -0.001) and highly overlapping for the observed stock returns estimates across all states/regimes.

5 Application

In this section, we use the different regime switching models in a forecasting exercise. This application allows us to explore the practical implications (forecasting performance in this case) of an improved regime switching model. More specifically, we attempt to forecast the future equity risk premium using a step-wise model selection approach under different equity market regimes, following Guidolin and Pedio [12]. We compare the forecasts obtained using

different regime indicators (horizon spread, stock index returns and conditional volatility) to determine if improved regime detection can also improve forecasting performance.

We build on the benchmark model of Campbell and Thompson [6] and predict the equity premium at time t as a linear function of the index dividend yield at time $t-1$. Further, we introduce regime dependency and write:

$$r_{s,t} = \alpha_s + \beta DY_{s,t-1} + \epsilon_{s,t} \tag{2}$$

where $r_{s,t}$ is the equity risk premium for time t under regime s, and $DY_{s,t-1}$ is the dividend yield for time $t-1$.

To compare the forecasting performance obtained with different regime switching models, we split our data into two periods. We use the period from January 2006 to December 2018 for estimation of the models, and the period from January 2019 to August 2020 to evaluate their out-of-sample forecasting performance. The split of the data into a period for estimation and another period for out-of-sample evaluation is chosen to allow for (i) a sufficiently long period to calibrate the models, and (ii) the inclusion of a crisis period in both samples: the financial crisis in the calibration period, and the Covid pandemic in the out-of-sample evaluation period.

Using the calibration sample, we run predictive regressions according to equation 2. Based on our setting in section 2.2, we run two predictive regressions of the equity risk premium: one for the expansion regime of the stock market if the probability of expansion regime exceeds 0.5, and the second for the contraction regime when the probability of the expansion regime is less than 0.5.

We then form a one-step ahead forecast of the equity risk premium. This is done out-of-sample, in a recursive fashion. At each time t in the out-of-sample period, we estimate the filtered probability of an expansion/contraction regime for time $t+1$ (based on the data observed up to time t). Using the estimated probability, we then compute the probability-

weighted average of the expansion and contraction predictions from equation 2 as the risk premium forecast for time $t+1$. We repeat this procedure for the three regime estimation models: horizon spread, stock index returns and conditional volatility. As a benchmark, we also form forecasts using equation 2 without any regime considerations.

To evaluate the forecast performance of the benchmark model as well as the three different regime switching models, we compute the RMSFE (root mean square forecast error) for the models, defined as:

$$RMSFE = \sqrt{\frac{\sum_{t=1}^{T} (\hat{y}_t - y_t)^2}{T}}$$

The RMSFE computed for the different models during the out-of-sample period is reported in Table 4. We can see that for both the US and the emerging markets, the horizon spread model has the lowest RMSFE, implying that it outperforms the benchmark and the other regime switching models in terms of forecasting performance. In addition, we note that forecast performance estimated using the other regime switching models (stock index returns and conditional volatility) is also better than the benchmark model in both markets, although the improvement is marginal.

Overall, our results show that option prices provide useful information about equity market regimes. Compared to using historical returns, forward-looking information extracted from option prices improves both regime detection and equity premium prediction.

Table 4: Forecast Performance of the Alternative Regime Models

The table below presents the root mean square forecast error (RMSFE) for the prediction of the equity risk premium based on the alternative regime switching models.

	No Regime	Horizon Spread	Stock Index Returns	Conditional Volatility
United States	0.01697	0.01691	0.01696	0.01696
Emerging Markets	0.01775	0.01771	0.01774	0.01773

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